

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester IV Examination, 2013-14

Paper No. BSC-13A: Statistical Inference, Sampling and Design of Experiments

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1.

- (a) Define probability. State and prove the addition theorem of probability for any two events A and B , i.e.,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Also state the theorem if A and B are independent events. [7]

- (b) A balanced die is thrown two times. Find the probability that the sum of the numbers in the two throws is:

- (i) 7
- (ii) more than 10
- (iii) less than 5

[7]

Question 2.

- (a) Define a discrete random variable and its probability mass function (p.m.f.) giving a suitable example. [7]
- (b) Explain the binomial and Poisson distributions. Discuss their properties and mention fields of their applications. [7]

Question 3.

- (a) Explain the probability density function (p.d.f.) and cumulative distribution function (c.d.f.) of a continuous random variable. [7]
- (b) Check whether the function defined as:

$$f(x) = \begin{cases} e^{-x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

is a probability density function. [7]

Question 4.

- (a) State the important properties of a normal distribution. [7]
 - (b) Explain the concept of testing of hypothesis. In this context, explain two types of errors (Type I and Type II errors) and the level of significance. [7]
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Question 5.

- (a) Explain the analysis of variance (ANOVA) table for a one-way classified data (under Completely Randomized Design). [7]
 - (b) Explain two uses of the χ^2 -test and two uses of the t -test. [7]
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Question 6.

- (a) Explain the terms: population, sample, and sampling frame giving suitable examples. [7]
 - (b) Describe simple random sampling. Explain a method to select a simple random sample from a population. [7]
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Question 7. Describe stratified random sampling. Derive the variance of the stratified sample mean estimator and explain proportional and optimum (Neyman) allocations. [14]

Question 8. Write short notes on any **three** of the following:

- (a) Conditional probability
- (b) Statistical definition of probability
- (c) Mean of binomial and Poisson distributions
- (d) Large sample test for single mean with one example
- (e) Advantages of sampling over complete enumeration

[14]